UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Artificial Intelligence

Course Code : **SAIA 2113**

Course Name : **Machine Learning**

Type of Assessment : Report & Presentation

Full Marks : 40%

Date : 9 December 2025 (Tuesday)

This group assignment consisted of 4 parts. Answer for all questions.

Name & Matric ID:	1. 2. 3. 4.
Section:	
Lecturer's Name:	

Report & Presentation

- Report : 30%
- Presentation : 10%

  Total : 40%

Submission Guidelines:

- **Report** in PDF format, submitted via e-Learning.
- **Team size**: 4 students per group.
- **Turnitin**: ≤ 25 percent. **Report submission did not include the questions and rubrics**.
- **Submission includes the coding file in .**ipynb.
- **Report Submission Dateline**: 8 January 2026, Khamis (11:59pm).
- **Presentation (Week 15)**: 13 January 2026, Tuesday (12 pm – 12:50 pm) & 14 January 2026, Wednesday (11 am – 1 pm).
- **Presentation 10 min, Q&A session 5 min**.

Assignment Question:

1. You are required to **build a machine learning application** using a suitable **machine learning framework (Artificial Neural Networks)**.
2. You must also **demonstrate ethical considerations** in your approach such as data privacy, fairness, and transparency.

Project Requirements:

1. **Select a real-world problem** in one of the following domains:
   - Healthcare
   - Agriculture
   - Finance
   - Smart City/Transportation
   - Environmental Monitoring

2. **Build a machine learning framework** that involves:
    - A publicly available dataset size in the range of 500 – 1k.
    - Perform data cleaning and EDA.
    - Number of features (≥ 10 features).
    - Model evaluation (i.e. Accuracy, precision, recall, F1- score and confusion matrix).

3. **Prepare a detailed project report** that includes the following sections:
    - Summary
    - Introduction
        - Problem Background
        - Aim
    - Literature Review (≥ 5 references within year 2022 – 2025)
    - Methodology
    - Results and Discussion
    - Conclusion
    - References

4. **Deliver a 10-minute group presentation** to showcase your project, including:
    - Problem statement
    - Dataset used
    - Model architecture
    - Results
    - Ethical considerations

Assessment Rubrics

Project Report (30%)

Criteria	Excellent (5)	Good (4)	Fair (3)	Poor (2)	Very Poor (1)	Marks
1. Summary (2%)	Clear summary of the problem, aim, method, results, and conclusion.	Lack of one component of the abstract.	Lack of two components of the abstract.	Lack of three components of the abstract.	None/Lack of four components of the abstract.	/5
2. Introduction (3%)	Clear problem background and aim.	Lack of explanation about the problem's background and aim.	Have discussion but lack one component.	Have discussion but lack two components.	None.	/5
3. Literature Review (4%)	More than or equal to 5.	Less than 5.	Less than 4.	Less than 3.	Less than 2 / None.	/5
4. Methodology P1 (4%)	A publicly available dataset size in the 500 – 1k range, Perform data cleaning and EDA, Number of features (≥ 10 features), Use of algorithms: Artificial Neural networks., Model evaluation (i.e. Accuracy, precision, recall, F1- score and confusion matrix).	Lack of one component.	Lack of two components.	Lack of three components.	None/Lack of four components.	/5
5. Methodology P2: ANN Architecture (4%)	≥ 95% Metric Performance.	71 - 94% Metric Performance.	61 - 70% Accuracy performance	≤60% Accuracy performance	Unsuccessful Run	/5
6. Results & Discussion (8%)	Clear discussion of the model evaluation (i.e. Accuracy, precision, recall, F1- score and confusion matrix).	Lack of one component.	Lack of two components.	Lack of three components.	None/Lack of four components.	/5
7. Ethical Consideration (3%)	Four ethical considerations.	Three ethical considerations.	Two ethical considerations.	One ethical consideration.	None.	/5
8. Report Structure, Formatting & Referencing (2%)	Well-organized, consistent citations, follow report format, Turnitin ≤ 25, and well-structured figures and tables.	Lack of one component.	Lack of two components.	Lack of three components.	None/Lack of four components.	/5

TOTAL						**/40 (scaled to 30%)**

Project Presentation (10%)

Criteria	Excellent (3)	Fair (2)	Poor (1)	Marks
Clarity of Problem Statement (2%)	Problem clearly stated and relevant	Problem stated but lack of explanation	Vague or unclear/None	/3
Methodology Explained (2%)	Fully understandable and well presented	Presented but lack of explanation	Unclear/None	/3
Results & Visuals (2%)	Fully understandable and well presented	Presented but lack of explanation	Unclear/None	/3
Ethical Reflection (2%)	Fully understandable and well presented	Presented but lack of explanation	Unclear/None	/3
Delivery & Communication (1%)	Confident, clear, professional	Lacks engagement or clarity	Poorly delivered	/3
Q&A Handling (1%)	Well-prepared, accurate answers	Hesitant or partial answers	Unable to respond	/3
TOTAL				**/18 scaled to 10%)**